

Exercice 1  
 $A = (01110111)_2$   $B = (00100000)_2$   $D = (10001001)_2$   $E = (11100000)_2$

2. Affichage sur Ecran (-105 et +105)

$$A = 01110111$$

$$B = 00100000$$

$$C = 10010111$$

$$C = -128 + 16 + 4 + 2 + 1 = -105$$

$$D = 10001001$$

$$E = 11100000$$

$$C = 01101001$$

$$C = -64 + 32 + 8 + 1 = +105$$

3. Biais =  $2^{N-1} - 1 = 2^{6-1} - 1 = 31$

4.

$$X = 96.125 = (1100000,001)_2 = (1,100000001)_2 \times 2^6$$

$$ED = ER + 31 = 6 + 31 = +37$$

$$X = (0\ 100101\ 1000000010000)_2$$

$$Y = X/32 = (1100000,001)_2 \times 2^{-5} = (1,100000001)_2 \times 2^{-1}$$

$$ED = ER + 31 = 1 + 31 = +32$$

$$Y = (0\ 100000\ 1000000010000)_2$$

$$Z = -28.8 = (11100,11001100\dots)_2 = -(1,110011001100\dots)_2 \times 2^4$$

$$ED = ER + 31 = 4 + 31 = +35$$

$$Z = (1\ 100011\ 1100110011001)_2$$

5. L'intervalle

$$[-(2^{+14} - 1) \cdot 2^{+18}, -2^{-30}] \cup [+2^{-30}, +(2^{+14} - 1) \cdot 2^{+18}]$$

Exercice 2

$$1) \overline{A}C + \overline{C}\overline{D} + \overline{A}B\overline{D} = \overline{A}C + \overline{C}\overline{D}$$

$$\overline{A}C + \overline{C}\overline{D} + \overline{A}B\overline{D} \cdot (C + \overline{C}) =$$

$$\overline{A}C + \overline{C}\overline{D} + \overline{A}B\overline{D} \cdot C + \overline{A}B\overline{D} \cdot \overline{C} =$$

$$\overline{A}C(1 + B\overline{D}) + \overline{C}\overline{D}(1 + \overline{A}B) = \overline{A}C + \overline{C}\overline{D}$$

# Corrigé Type Structure 2016

$$2) (\overline{A.B.C} + \overline{A.B.C})(A + A.B)(C + C.D)(\overline{E} + \overline{E.D}) + A.C\overline{E}.D =$$

selon Demorgan

$$(\overline{A.B.C} + \overline{A.B.C}) = \overline{A.B.C.A.B.C} = \overline{0} = 1$$

donc

$$(\overline{A.B.C} + \overline{A.B.C})(A + A.B)(C + C.D)(\overline{E} + \overline{E.D}) + A.C\overline{E}.D = (A + A.B)(C + C.D)(\overline{E} + \overline{E.D}) + A.C\overline{E}.D$$

selon la règle  $X + X.Y = X$  alors

$$(A + A.B)(C + C.D)(\overline{E} + \overline{E.D}) + A.C\overline{E}.D = A.C\overline{E} + A.C\overline{E}.D = A.C\overline{E}$$

$$3) \overline{A \oplus B} + \overline{A} + \overline{B} = A \otimes B + \overline{A.B} = A.B + \overline{A.B} + \overline{A.B} = A.B + \overline{A.B} = A \otimes B$$

$$4) (C + \overline{A.D} + \overline{B.C})(\overline{B.C.D.A}) = (C + \overline{A.D} + \overline{B.C})(\overline{B.C.C.D.A})$$

$$= (C + \overline{A.D} + \overline{B.C})(\overline{C.A.D.B.C}) = \underbrace{(C + \overline{A.D} + \overline{B.C})}_X \cdot \underbrace{(\overline{C.A.D.B.C})}_{\overline{X}}$$

## Exercice 3

| A | B | C | D | R | V | O |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 |



formes numériques

$$R = \sum (3, 6, 12)$$

$$V = \sum (5, 9, 10)$$

$$O = \sum (6)$$

$$O = \overline{A.B.C.D} = \overline{A.B.C.D} = \overline{A+B+C+D} = \overline{A+B+C+D} = \overline{A+B+B+C+C+D}$$